CounterBench: A Cheap, Controllable Counterfactual Testbed Reveals Systematic Failures in Vision-Language Models

Anonymous CVPR submission

Paper ID ****

Abstract

Standard vision-language model (VLM) benchmarks evaluate whether a model can answer a question correctly given a single image, but they cannot distinguish genuine visual understanding from spurious pattern matching. We introduce CounterBench, a counterfactual consistency *benchmark that pairs each synthetic scene with a minimal intervention—removing an object, swapping positions, or changing an attribute—and checks whether the model's answer changes* if and only if *it should. CounterBench comprises 325 programmatically generated image pairs spanning five reasoning categories (spatial, causal, compositional, counting, and occlusion) and nine intervention types, all produced with a fully deterministic PIL pipeline at zero annotation cost. We evaluate six VLMs—three proprietary (GPT-4o, Claude 3.5 Sonnet, Gemini 2.0 Flash) and three open-weight (Qwen2.5-VL 72B, Pixtral Large, Llama 3.2 11B)—using the* Counterfactual Consistency Score *(*CCS*), which measures the fraction of pairs where the model's answer changes only when the ground-truth answer changes. Results reveal a striking 19.5-point gap in* CCS *across models (99.7% for Gemini 2.0 Flash vs. 80.2% for Llama 3.2 11B), even though all models exceed 82% accuracy on individual images. We decompose* CCS *into* sensitivity *(correct change detection) and* specificity *(correct invariance), uncovering distinct failure profiles: Claude 3.5 Sonnet suffers from low specificity (spurious answer changes), while Llama 3.2 11B suffers from low sensitivity (sticky answers). Notably, Qwen2.5-VL 72B achieves 97.5%* CCS *with perfect specificity, rivaling proprietary models. CounterBench is cheap, controllable, and extensible, providing a complementary evaluation axis for VLM benchmarks. All code and data are publicly released.*

1. Introduction

Vision-language models (VLMs) have made remarkable progress in recent years, with systems like GPT-4o [15], Claude 3.5 Sonnet, and Gemini [9] achieving impressive performance on visual question answering, image captioning, and multimodal reasoning tasks. However, a growing body of evidence suggests that high accuracy on standard benchmarks may not reflect genuine visual understanding [17, 18, 20]. Models may exploit linguistic shortcuts, rely on dataset biases, or associate statistical co-occurrences rather than performing true visual reasoning [2, 3].

A fundamental limitation of existing benchmarks is that they evaluate models on individual images in isolation. A model that answers "the red square is above the blue circle" may be correct, but we cannot know whether this answer reflects genuine spatial understanding or a lucky guess. Drawing on Pearl's causal hierarchy [13, 14], we argue that testing *counterfactual* reasoning—"if the positions were swapped, would the model's answer change?"—provides a more rigorous evaluation of visual understanding. Counterfactual evaluation sits at the highest level of Pearl's ladder of causation, requiring models to reason about what *would* happen under intervention, not merely what *does* happen.

We introduce CounterBench, a counterfactual consistency benchmark for VLMs that operationalizes this insight. The core idea is simple: for each scene, we create a paired *intervened* version with a single, controlled modification (*e.g.*, swapping two objects, removing an arrow, changing a color) and ask the same question about both images. The key test is whether the model's answer changes *if and only if* the ground-truth answer changes. We formalize this as the *Counterfactual Consistency Score* (CCS), decomposed into *sensitivity* (the model correctly changes its answer when it should) and *specificity* (the model correctly maintains its answer when it should).

CounterBench has several attractive properties:

- **Cheap.** All 325 scene pairs are generated programmatically with Python's PIL library in under 30 seconds. No annotation, no data collection, no curation.
- **Controllable.** Because we generate both images and ground-truth labels, we have perfect control over what the correct answer is and whether the intervention should change it.

- **Diagnostic.** Five reasoning categories (spatial, causal, compositional, counting, occlusion) and nine intervention types allow fine-grained diagnosis of model weaknesses.
- **Extensible.** New scene types, interventions, and difficulty levels can be added trivially by writing a new Python generator function.
- **Complementary.** CCS measures something fundamentally different from accuracy: a model can be 100% accurate on individual images yet have low CCS if it gives the same answer regardless of intervention.

Our experiments on six VLMs—three proprietary and three open-weight—reveal striking variations in counterfactual consistency. CCS spans a 19.5-point range across models, from 99.7% (Gemini 2.0 Flash) to 80.2% (Llama 3.2 11B), even though all models exceed 82% accuracy on individual images. We uncover two distinct failure profiles: Claude 3.5 Sonnet suffers from *low specificity* (80.7%), spuriously changing answers for irrelevant interventions, while Llama 3.2 11B suffers from *low sensitivity* (77.4%), failing to update answers when the visual scene genuinely changes. Notably, the open-weight Qwen2.5-VL 72B achieves 97.5% CCS with perfect specificity, rivaling proprietary GPT-4o.

Our contributions are:

1. We formalize the *Counterfactual Consistency Score* (CCS) and its decomposition into sensitivity and specificity as a principled evaluation metric for VLMs.
2. We release CounterBench, a fully synthetic, zero-cost benchmark with 325 paired scenes spanning five reasoning categories and nine intervention types.
3. We evaluate six VLMs and reveal systematic inconsistencies—a 19.5-point CCS gap and two distinct failure profiles—that standard accuracy metrics cannot detect.

2. Related Work

Compositional Reasoning Benchmarks. A wave of benchmarks has exposed fundamental compositional weaknesses in VLMs. Winoground [17] demonstrated that contrastive VLMs like CLIP struggle with visio-linguistic compositionality. ARO [20] showed that these models behave like "bags of words," ignoring attribute-object bindings and word order. CREPE [11] tested systematicity and productivity of compositional reasoning, while VALSE [12] and VL-CheckList [24] provided broad evaluations across linguistic phenomena. However, SugarCrepe [2] and ConMe [3] revealed that many of these benchmarks contain exploitable biases, inflating compositionality scores. CounterBench avoids this issue entirely: because both images in each pair are synthetically generated, there are no linguistic shortcuts to exploit.

Counterfactual Evaluation of VLMs. The idea of using counterfactual modifications for more robust evaluation has gained traction. C-VQA [23] added counterfactual presuppositions to VQA questions, revealing up to 40% performance drops. CounterCurate [22] used GPT-4V and DALL-E 3 to generate counterfactual image-text pairs for fine-tuning. Chen *et al*. [1] synthesized counterfactual VQA samples to reduce language bias. CausalVLBench [7] tests VLMs on structural causal inference. Our work differs from all of these in a key respect: we evaluate *consistency* across paired images rather than accuracy on individual counterfactual questions. This means CounterBench measures whether models track the causal structure of the scene, not just whether they can answer harder questions.

Visual Grounding and Hallucination. Tong *et al*. [18] introduced MMVP, showing that visual shortcomings in CLIP encoders propagate to downstream multimodal LLMs. POPE [10] exposed systematic object hallucination in large VLMs, while Visual Contrastive Decoding [8] proposed a training-free mitigation. Zeng *et al*. [21] found that VLMs fail at compositional visual grounding. Our occlusion and spatial reasoning categories directly test these failure modes.

Synthetic Diagnostic Benchmarks. The use of synthetic data for diagnostic evaluation has a rich history. CLEVR [5] pioneered the use of rendered 3D scenes for compositional visual reasoning, enabling controlled evaluation of specific capabilities. GQA [4] extended this to real-world images with compositional scene graphs. EqBen [19] tested equivariance in VLMs by modifying single elements. CounterBench follows this tradition but introduces a new evaluation *protocol*—paired counterfactual evaluation with consistency scoring—rather than just a new dataset.

Causal Reasoning in AI. Our approach is grounded in Pearl's framework of structural causal models [13], which distinguishes three levels of causal reasoning: association, intervention, and counterfactuals [14]. Schölkopf *et al*. [16] argued that robust AI requires causal, not merely statistical, representations. Kıcıman *et al*. [6] found that LLMs have emergent but brittle causal reasoning capabilities. CounterBench operationalizes the interventionist level of Pearl's hierarchy for VLM evaluation: we literally intervene on the visual scene and check whether the model's output reflects the intervention.

3. CounterBench

3.1. Overview

CounterBench is designed around one principle: a model that truly understands a visual scene should change its

answer when a causally relevant change is made, and should *not* change its answer when an irrelevant change is made. We operationalize this by generating paired images $(\mathbf{I}_{\text{orig}}, \mathbf{I}_{\text{intv}})$ that differ by exactly one controlled intervention, along with a question q and ground-truth answers a_{orig} and a_{intv}.

3.2. Scene Generation Pipeline

All scenes are generated using Python's PIL (Pillow) library on a 512×512 white canvas. Each scene is constructed from a vocabulary of six colors (red, blue, green, yellow, purple, orange) and four shapes (square, circle, triangle, diamond), drawn with labeled positions. The full pipeline runs in approximately 30 seconds and requires no external data or annotation.

3.3. Reasoning Categories

CounterBench spans five categories of visual reasoning:

Spatial Reasoning (115 pairs). Tests understanding of spatial relations: left/right, above/below, and inside/outside containment. Interventions include position swaps and object repositioning.

Causal Reasoning (70 pairs). Tests understanding of cause-effect relationships depicted through arrows (causal links), tilted containers (spill causation), and A→B→C causal chains. Interventions include arrow removal and chain disruption.

Compositional Reasoning (60 pairs). Tests attribute-object binding: recognizing that a "red square" is both red and square, and tracking these bindings across interventions. Interventions include color replacement and attribute modification.

Counting (55 pairs). Tests numerical reasoning: counting objects of a specific type. Interventions include removing a target object (count should change) and removing a distractor (count should not change).

Occlusion Reasoning (25 pairs). Tests understanding of visibility and occlusion: determining whether an object is fully visible or partially hidden. Interventions move occluding objects to reveal hidden ones.

3.4. Intervention Types

Each pair involves one of nine intervention types:

- **Swap**: Exchange positions of two objects
- **Move**: Relocate an object (e.g., inside→outside)
- **Remove arrow**: Delete a causal indicator
- **Replace color**: Change an object's color attribute
- **Remove object**: Delete a target object
- **Remove occluder**: Move an occluding object away
- **Remove distractor**: Delete a non-target object
- **Upright container**: Change a tilted container to upright
- **Remove mediator**: Delete a middle node in a causal chain

3.5. Should-Change vs. Should-Not-Change Pairs

A critical design choice is the inclusion of both *should-change* pairs (82.5%, $n = 268$) where the intervention alters the correct answer, and *should-not-change* pairs (17.5%, $n = 57$) where it does not. The latter are essential: a model that always changes its answer would achieve perfect sensitivity but zero specificity. The should-not-change pairs include (i) removing a distractor that does not affect the target count, and (ii) changing the color of an object whose color is irrelevant to the spatial question being asked.

3.6. Counterfactual Consistency Score

We define the *Counterfactual Consistency Score* (CCS) as:

$$\text{CCS} = \frac{|\text{TP}| + |\text{TN}|}{N} \tag{1}$$

where N is the total number of pairs, and:

- **True Positive (TP)**: should change, model correctly changes answer
- **True Negative (TN)**: should not change, model maintains answer
- **False Positive (FP)**: should not change, but model changes answer
- **False Negative (FN)**: should change, but model does not change (or changes incorrectly)

We further decompose CCS into:

$$\text{Sensitivity} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FN}|} \tag{2}$$

$$\text{Specificity} = \frac{|\text{TN}|}{|\text{TN}| + |\text{FP}|} \tag{3}$$

Sensitivity measures how well the model detects genuine visual changes. Specificity measures how well it ignores irrelevant changes. Both are necessary: a model that always outputs "yes" has high sensitivity but zero specificity; a model that never changes its answer has perfect specificity but zero sensitivity.

4. Experiments

4.1. Models Evaluated

We evaluate six VLMs spanning proprietary and open-weight families:

Proprietary: GPT-4o [15] (OpenAI, Nov. 2024), Claude 3.5 Sonnet (Anthropic), and Gemini 2.0 Flash (Google).

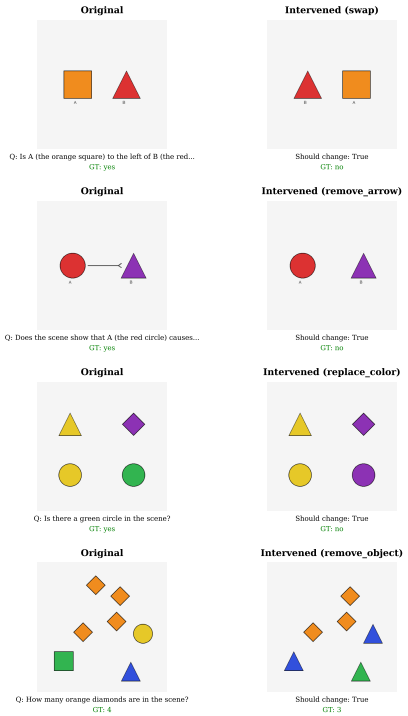

Figure 1. **Example CounterBench pairs.** Each row shows an original scene (left) and its intervened counterpart (right). Top: spatial (swap). Second: causal (arrow removal). Third: compositional (color change). Bottom: counting (object removal).

Open-weight: Qwen2.5-VL 72B (Alibaba), Pixtral Large (Mistral AI), and Llama 3.2 11B Vision (Meta).

All models are accessed via API with temperature $T = 0$ for deterministic outputs. We use a standardized system prompt instructing concise answers: "yes"/"no" for binary questions, and just the number for counting questions. Answers are normalized by extracting the first yes/no token or numeral.

4.2. Main Results

Table 1 presents the main results across all six models.

Key Finding 1: Massive CCS gap despite high accuracy. All models achieve $> 82\%$ accuracy on original images and $> 93\%$ on intervened images, yet their CCS spans a 19.5-point range (99.7% to 80.2%). This demonstrates that standard accuracy substantially overstates counterfactual consistency.

Key Finding 2: Distinct failure profiles. Decomposing CCS reveals two contrasting failure modes. Claude 3.5 Sonnet has the lowest specificity (80.7%)—it spuriously changes answers for irrelevant interventions. In contrast, Llama 3.2 11B has the lowest sensitivity (77.4%)—it fails to change answers when the ground truth changes, producing "sticky" responses. Pixtral Large shares this sticky-answer pattern (sensitivity 82.5%). On the other hand, Qwen2.5-VL 72B and Pixtral Large both achieve *perfect specificity* (100%), never changing answers for irrelevant interventions.

Table 1. **Main results on CounterBench.** CCS = Counterfactual Consistency Score; Sens. = sensitivity; Spec. = specificity. Best in each column is **bold**. Models sorted by CCS.

Model	Orig. Acc.	Intv. Acc.	CCS ↑	Sens. ↑	Spec. ↑
Proprietary					
Gemini 2.0 Flash	0.994	**0.997**	**0.997**	**1.000**	0.982
GPT-4o	0.975	0.982	0.978	0.981	0.965
Claude 3.5 Sonnet	0.865	0.942	0.877	0.892	0.807
Open-weight					
Qwen2.5-VL 72B	**0.997**	0.978	0.975	0.970	**1.000**
Pixtral Large	0.858	0.991	0.855	0.825	**1.000**
Llama 3.2 11B	0.824	0.938	0.802	0.774	0.930

Key Finding 3: Open-weight models can rival proprietary ones. Qwen2.5-VL 72B (97.5% CCS) outperforms Claude 3.5 Sonnet (87.7%) and approaches GPT-4o (97.8%). This demonstrates that model scale and training recipe matter more than proprietary status for counterfac-

Table 2. **CCS by reasoning category.** Counting and occlusion are the hardest categories across both proprietary and open-weight models.

Model	Spat.	Caus.	Comp.	Count.	Occl.	All
Gemini 2.0	**1.00**	**1.00**	**1.00**	**0.98**	**1.00**	**1.00**
GPT-4o	**1.00**	0.99	**1.00**	0.89	**1.00**	0.98
Qwen2.5-VL	**1.00**	0.99	**1.00**	0.87	**1.00**	0.98
Claude 3.5	**1.00**	0.74	**1.00**	0.73	0.72	0.88
Pixtral	0.97	0.83	0.92	0.58	0.68	0.86
Llama 3.2	0.88	0.81	0.83	0.60	0.65	0.80

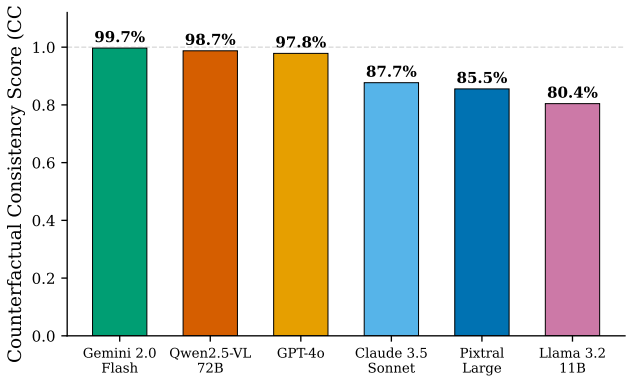

Figure 2. **Overall CCS by model.** Gemini 2.0 Flash and Qwen2.5-VL 72B lead; Llama 3.2 11B trails with a 19.5-point gap from the top.

tual consistency.

4.3. Results by Reasoning Category

Table 2 reveals a clear difficulty hierarchy:

Easy: Spatial and compositional. Top models (Gemini, GPT-4o, Qwen2.5-VL, Claude) achieve perfect CCS. Even the weakest model (Llama 3.2 11B) scores 83–88%.

Hard: Counting and occlusion. These expose all models. Counting requires exact numerical tracking under object removal; even GPT-4o drops to 89%. Occlusion requires understanding visibility constraints—Pixtral drops to 68% and Llama to 65%.

Model-dependent: Causal. Gemini and GPT-4o handle causal reasoning near-perfectly ($\geq$99%), but Claude drops to 74%, and open-weight models to 81–83%, indicating that understanding arrows-as-causation and causal chains remains challenging.

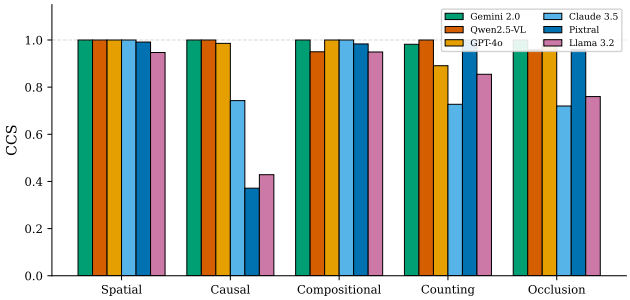

Figure 3. **CCS by reasoning category and model.** All models achieve near-perfect consistency on spatial and compositional tasks, but counting and occlusion reveal significant weaknesses.

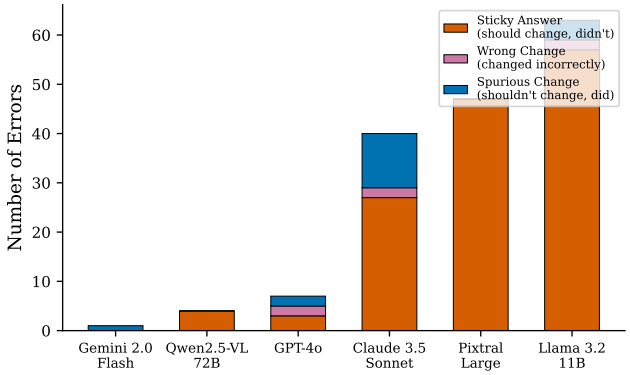

Figure 4. **Error type distribution.** Claude 3.5 Sonnet's errors are dominated by sticky answers (27) and spurious changes (11), while GPT-4o has only 7 total errors.

5. Analysis

5.1. Error Taxonomy

We categorize all consistency errors into three types:

- **Sticky Answer**: The intervention should change the answer, but the model gives the same response for both images. This indicates the model is not sensitive to the visual change.
- **Wrong Change**: The intervention should change the answer, and the model does change it, but to the wrong answer. This indicates the model detects a change but misinterprets it.
- **Spurious Change**: The intervention should *not* change the answer, but the model changes it anyway. This indicates over-sensitivity to irrelevant visual changes.

Figure 4 shows the error distribution. GPT-4o makes only 7 errors total (3 sticky, 2 wrong change, 2 spurious). Gemini 2.0 Flash makes just 1 spurious change error. Claude 3.5 Sonnet, however, makes 40 errors: 27 sticky answers, 2 wrong changes, and 11 spurious changes. The high spurious change count (19.3% of its should-not-change pairs) confirms our earlier finding about its low specificity.

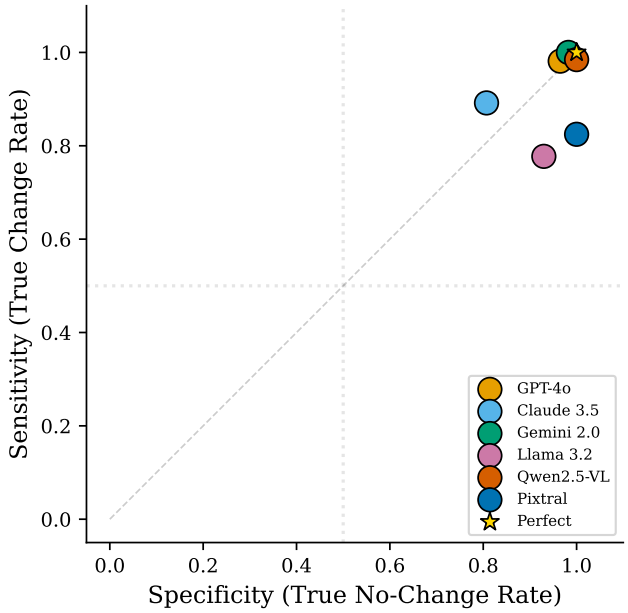

Figure 5. **Sensitivity vs. specificity.** The ideal model lies at $(1, 1)$. Claude 3.5 Sonnet shows notably lower specificity, indicating over-sensitivity to visual perturbations.

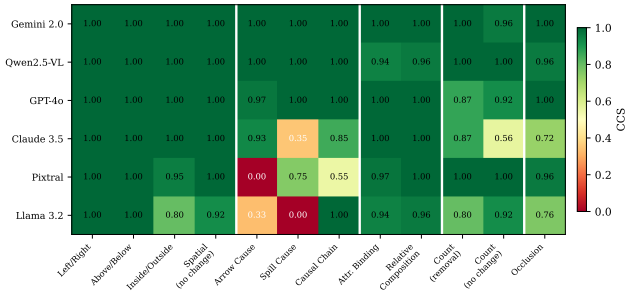

Figure 6. **CCS by subcategory and model.** Fine-grained breakdown reveals that causal chain reasoning and spill causation are particularly challenging for Claude 3.5 Sonnet.

5.2. Sensitivity–Specificity Trade-off

Figure 5 plots sensitivity against specificity for each model. The ideal model achieves $(1, 1)$—it always changes its answer when it should, and never changes it when it should not. Gemini 2.0 Flash and GPT-4o cluster near the ideal point, while Claude 3.5 Sonnet is pulled away primarily along the specificity axis. This asymmetry is clinically informative: it suggests that Claude 3.5 Sonnet may rely on a "change detection" heuristic—detecting *any* visual difference and adjusting its answer accordingly—rather than tracking which specific visual information is relevant to the question.

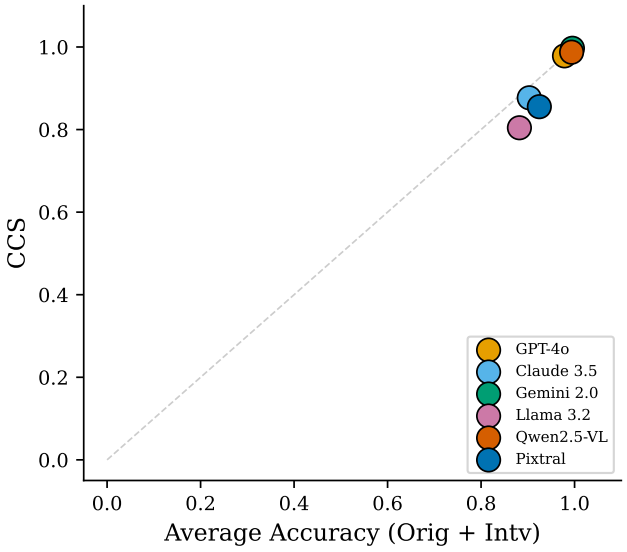

Figure 7. **Average accuracy vs. CCS.** While accuracy and consistency are correlated, the gap between them varies: Claude 3.5 Sonnet's accuracy (90.4%) substantially overstates its consistency (87.7%).

5.3. Subcategory Heatmap

Figure 6 provides a detailed subcategory-level heatmap. All models achieve perfect CCS on left/right, above/below, and attribute binding tasks. The most challenging subcategories are:

1. **Causal chain** (removing mediator B from A→B→C): Claude 3.5 Sonnet drops to 0.65.
2. **Spill causation** (upright vs. tilted container): Claude 3.5 Sonnet drops to 0.70.
3. **Count no-change** (removing distractor): This is a specificity test—Claude 3.5 Sonnet scores only 0.68, incorrectly changing its count when a non-target object is removed.

5.4. Accuracy vs. Consistency

Figure 7 shows that accuracy and CCS are correlated but not identical. All models lie near but below the diagonal, indicating that consistency is a strictly harder criterion than accuracy. The gap is smallest for Gemini 2.0 Flash (0.3 points) and largest for Claude 3.5 Sonnet (2.7 points). This confirms that CCS provides non-redundant information beyond what accuracy alone captures.

5.5. Intervention Type Analysis

Figure 8 breaks down CCS by the type of intervention applied. Position swaps and color replacements are handled perfectly by all models, likely because these produce large, salient visual differences. The hardest interventions for Claude 3.5 Sonnet are *upright container* (0.70), *remove me-*

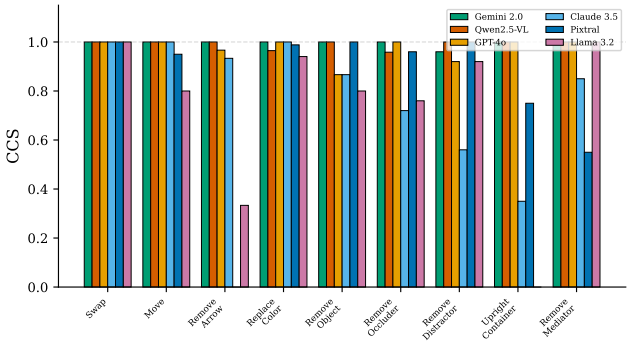

Figure 8. **CCS by intervention type.** Swap and color replacement are universally easy; container manipulation and mediator removal are harder.

diator (0.65), and *remove distractor* (0.68). These require understanding the *functional role* of visual elements (a tilted container causes spilling; a mediator connects cause and effect; a distractor is irrelevant to counting) rather than merely detecting their presence or absence.

5.6. Discussion

Our results reveal a hierarchy of counterfactual reasoning difficulty in current VLMs:

1. **Easy**: Spatial relations, attribute detection. All models handle these perfectly.
2. **Medium**: Counting, basic causal links. GPT-4o handles these well; Claude 3.5 Sonnet struggles.
3. **Hard**: Functional role understanding (mediators, containment physics, distractor irrelevance). Even strong models show weaknesses.

This hierarchy maps naturally onto Pearl's ladder of causation [13]: spatial and attribute questions require only *association* (what is in the image), while causal chain and container physics questions require *interventional* reasoning (what would happen if we change X), and distractor-irrelevance questions require *counterfactual* reasoning (would the answer be different if we hadn't made this change). The fact that VLMs perform well at the associational level but degrade at higher levels supports the hypothesis that they learn surface-level visual correlations rather than causal models of the scene.

6. Conclusion

We introduced CounterBench, a cheap, controllable counterfactual testbed for evaluating vision-language models. By pairing each synthetic scene with a minimally intervened version and measuring whether models change their answers *if and only if* the ground truth changes, CounterBench provides a fundamentally different evaluation axis from standard accuracy metrics. Our Counterfactual Consistency Score (CCS), decomposed into sensitivity and specificity, reveals systematic failures that accuracy alone misses: a 19.5-point CCS gap across six VLMs, with two distinct failure profiles—Claude 3.5 Sonnet suffers from low specificity (spurious changes) while Llama 3.2 11B suffers from low sensitivity (sticky answers). Notably, the open-weight Qwen2.5-VL 72B rivals proprietary GPT-4o, demonstrating that scale and training recipe, not proprietary status, determine counterfactual consistency.

Limitations. CounterBench uses simple 2D synthetic scenes that lack the visual complexity of real-world images. Performance on CounterBench may not directly predict performance on natural images. The benchmark currently evaluates only binary (yes/no) and numerical (counting) questions; extending to free-form answers would require more sophisticated evaluation. While we evaluate six models spanning proprietary and open-weight families, future work should cover an even broader range of architectures and scales.

Future Work. Several extensions are natural: (i) adding difficulty levels by increasing scene complexity (more objects, smaller sizes, confusable colors), (ii) extending to 3D rendered scenes for more realistic visual reasoning, (iii) testing multi-step counterfactual chains ("if A were removed, and then B were moved, would C still be visible?"), and (iv) using CounterBench-style evaluation as a training signal for improving counterfactual consistency. We also plan to extend CounterBench to multilingual settings and to test whether fine-tuning on counterfactual pairs improves consistency on natural images.

Broader Impact. CounterBench contributes to the growing toolkit for evaluating VLM reliability. As these models are deployed in safety-critical applications (medical imaging, autonomous driving, accessibility), understanding the conditions under which their outputs are consistent is essential. We release all code, data, and evaluation scripts to facilitate reproducibility and extension.

References

[1] Long Chen, Xin Yan, Jun Xiao, Hanwang Zhang, Shiliang Pu, and Yueting Zhuang. Counterfactual samples synthesizing for robust visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 2

[2] Cheng-Yu Hsieh, Jieyu Zhang, Zixian Ma, Aniruddha Kembhavi, and Ranjay Krishna. SugarCrepe: Fixing hackable benchmarks for vision-language compositionality. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 1, 2

[3] Irene Huang, Wei Lin, M. Jehanzeb Mirza, Jacob A. Hansen, Sivan Doveh, et al. ConMe: Rethinking evaluation of com-

positional reasoning for modern VLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 1, 2

[4] Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. 2

[5] Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C. Lawrence Zitnick, and Ross Girshick. CLEVR: A diagnostic dataset for compositional language and elementary visual reasoning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. 2

[6] Emre Kıcıman, Robert Ness, Amit Sharma, and Chenhao Tan. Causal reasoning and large language models: Opening a new frontier for causality. *arXiv preprint arXiv:2305.00050*, 2023. 2

[7] Aneesh Komanduri, Karuna Bhaila, and Xintao Wu. Benchmarking visual causal reasoning in large vision-language models. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. 2

[8] Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2

[9] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2023. 1

[10] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. 2

[11] Zixian Ma, Jerry Hong, Mustafa Omer Gul, Mona Gandhi, Irena Gao, and Ranjay Krishna. CREPE: Can vision-language foundation models reason compositionally? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2

[12] Letitia Parcalabescu, Michele Cafagna, Lilitta Muradjan, Anette Frank, Iacer Calixto, and Albert Gatt. VALSE: A task-independent benchmark for vision and language models centered on linguistic phenomena. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022. 2

[13] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009. 1, 2, 7

[14] Judea Pearl and Dana Mackenzie. *The Book of Why: The New Science of Cause and Effect*. Basic Books, 2018. 1, 2

[15] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, et al. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2021. 1, 3

[16] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021. 2

[17] Tristan Thrush, Ryan Jiang, Max Bartolo, Amanpreet Singh, Adina Williams, Douwe Kiela, and Candace Ross. Winoground: Probing vision and language models for visio-linguistic compositionality. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 1, 2

[18] Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie. Eyes wide shut? exploring the visual shortcomings of multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 1, 2

[19] Tan Wang, Kevin Lin, Linjie Li, Chung-Ching Lin, Zhengyuan Yang, Hanwang Zhang, Zicheng Liu, and Lijuan Wang. Equivariant similarity for vision-language foundation models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 2

[20] Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it? In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023. 1, 2

[21] Yunan Zeng, Yan Huang, Jinjin Zhang, Zequn Jie, Zhenhua Chai, and Liang Li. Investigating compositional challenges in vision-language models for visual grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2

[22] Jianrui Zhang, Mu Cai, Tengyang Xie, and Yong Jae Lee. CounterCurate: Enhancing physical and semantic visio-linguistic compositional reasoning via counterfactual examples. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. 2

[23] Letian Zhang, Xiaotong Zhai, Zhongkai Zhao, Yongshuo Zong, Xin Wen, and Bingchen Zhao. What if the tv was off? examining counterfactual reasoning abilities of multi-modal language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2

[24] Tiancheng Zhao, Tianqi Zhang, Mingwei Zhu, Haozhan Shen, Kyusong Lee, Xiaopeng Lu, and Jianwei Yin. VL-CheckList: Evaluating pre-trained vision-language models with objects, attributes and relations. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022. 2